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Aerial image recognition in discriminative bi-transformer

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ABSTRACT

Aerial image recognition (AIR) has received increasing interest in the remote sensing community, and complex backgrounds with various types of objects bring huge challenges to recognition. Vision transformer (ViT) can capture long-range contextual relations, but it tends to fall into over-fitting for training samples. This problem is particularly acute in AIR where trainable data is insufficient. While training ViT with label smoothing can alleviate this problem, it is easy to pose the risk of underfitting and relies heavily on data quality. Furthermore, the application of ViT in AIR has always been limited because aerial images generally have greater intra-class variation and inter-class similarity than natural images. Thus, we propose a ViT-based framework called D-BiT to extend ViT to learn discriminative features in aerial images by integrating two heads on the ViT backbone. One is a classifier head with the Discriminative Label Smoothing Module (DLSM) to fully learn effective features without overfitting the distribution of noisy data, even in the condition of insufficient data volume. Another is a projector head with a newly-designed supervised contrastive loss to effectively combine label information and achieve a more compact and reasonable intra-class structure. Experimental results on three popular AIR benchmarks demonstrate the effectiveness of D-BiT: Aerial Image Dataset (AID), UC Merced and Event Recognition in Aerial videos (ERA), where we achieve state-of-the-art (SOTA) performance. In particular, D-BiT can significantly improve the accuracy of ViT for recognizing aerial images.

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1. Introduction

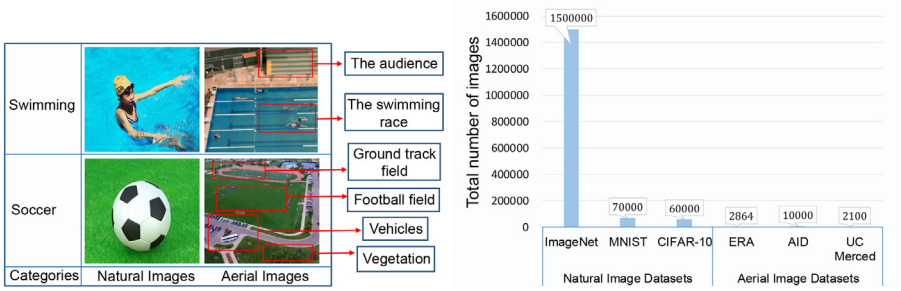
Aerial image recognition (AIR) is an active research topic in the intelligent interpretation of aerial images. AIR aims to distinguish a predefined semantic category (*e.g.*, "meadow", "resort", "railway station") for each given image according to its visual information [1–3]. Benefiting from the progress of deep neural networks, the existing convolutional neural network (CNN) models have shown excellent performance in AIR. Despite the advantages of exploiting the local features, the ability of CNN to model long-range relations within images is still insufficient. This is mainly due to the local receptive field of the convolution operator. Long-range spatial dependence is a common phenomenon in aerial images, characterized by their complex and compositional structure [1,4,5]. The discriminative features can be distributed in different regions of the aerial image. Therefore, modeling these global relations is crucial for understanding aerial imagery. Expanding the receptive field is the current common solution, but it requires more intensive yet damaging pooling operations [6,7].

Transformer-based models like vision transformer (ViT) [8] have gained increased popularity for image recognition tasks recently, signaling a transition from CNN-based models to ViT-based models. Briefly, ViT divides an image into a sequence of non-overlapping patches and then learns inter-patch representations using multi-headed self-attention in transformers [9]. This operation allows the model to aggregate rich global information, in stark contrast to the limited "receptive field" of convolutions. In addition, ViT is particularly effective at capturing spatial patterns and non-local dependencies. Yet, evidence suggests that ViT requires very large datasets to outperform the CNN, *e.g.* in [8], the benefits of ViT only became evident when Google's private 300 million image dataset, JFT-30M, was used for pre-training.

ViT's reliance on large-scale data is a huge obstacle to its widespread application, and it is easy to fall into overfitting for training samples, which greatly affects their generalization ability to test samples. As shown in Fig. 1b, this problem is particularly acute in AIR, where datasets are smaller and are often accompanied by less reliable labels. The standard solution is to employ transfer learning: Typically, a model is pre-trained on a larger-scale dataset such as ImageNet [10] and then fine-tuned for downstream tasks. Deep neural networks pre-trained on ImageNet generally

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(a) Comparison of natural images and aerial images in training samples.

(b) Comparison of natural images and aerial images in dataset size.

Fig. 1. Compared with natural images, aerial images possess complex backgrounds and various types of objects. At the same time, the datasets of aerial images are smaller, and the data available for training is generally insufficient.

outperform those trained from scratch in the remote sensing domain in terms of final performance and reduced training time [5,11,12]. However, there are still several issues:

Fitting dilemma. It is a common practice to represent a true label as a one-hot vector, during both the pre-training and fine-tuning stages. However, insufficient one-hot representations tend to train models with over-confidence. Label smoothing [13] can be used to remedy the inefficiency of one-hot vector labeling, but inevitably leads to model underfitting on small datasets. In nature, label smoothing encourages the representations of samples from the same class into tight clusters [14] by providing true labels with a certain degree of fault tolerance. But this also results in the loss of differences between these samples, posing the risk of model underfitting on small datasets. This dilemma is accentuated on aerial image datasets with noisy and difficult samples.

The dilemma of learning discriminative features. Aerial images usually have complex backgrounds and sparse object changes in the scene, so they have greater intra-class variation and inter-class similarity than natural images. And the simple cross-entropy loss in ViT is not enough to fully supervise the learning of discriminative semantic features. It is difficult for models to learn distinguishing features among different categories.

For the first problem, referring to the human learning process, *i.e.*, easy before difficult [15], a question arises: whether the fault tolerance of the model can be controlled by adaptive label smoothing during training? Therefore, in this paper, we propose the Discriminative Label Smoothing Module (DLSM) to guide the model that can start learning from easy samples and gradually progress to complex samples and knowledge. It mainly consists of a screening module of difficult samples and two smoothing variables. Obviously, the division of how easily samples are distinguished should be done according to how easily the model learns those samples. This can be achieved by the likelihood distribution of the model output. For example, the likelihood is mostly concentrated in the correct targets for simple samples and spread across multiple targets for difficult samples. In addition, to dynamically adjust the fault tolerance of the model, we replace the smoothing constant in label smoothing with two smoothing variables that control the smoothing weights for easy and hard samples, respectively. With the help of DLSM, ViT can spend less time on difficult samples in the early stage of training, and can gradually regress to the learning of these difficult samples in the later stage. This control is very useful for ViT to recognize aerial images because ViT can efficiently learn discriminative features while avoiding training around noisy data.

For the second problem, the family of contrastive objective functions that have flourished in self-supervised learning in recent years provides an idea. These functions can minimize the intra-class similarity and maximize inter-class differences, and also aim at modeling the intrinsic structure in data. However, the standard contrastive loss (*e.g.*, InfoNCE [16]) cannot be directly used for supervised learning, because it assumes that each sample in the minibatch belongs to a separate class. Khosla et al. [17] proposed a supervised contrastive loss to treat all samples with the same label in a minibatch as positive keys, adapting contrastive learning to the fully supervised setting. But aerial images have larger intra-class differences than natural images, and it is inappropriate to regard all samples in the same class as equal positive keys. To combat the above dilemmas, we propose a discriminative supervised contrastive loss that not only efficiently utilizes label information and discriminatively processes different augmented images in the same class, but also preserves the beneficial properties of contrastive objective functions.

Finally, combining the novel DLSM and discriminative supervised contrastive loss, we propose the final architecture called D-BiT to extend ViT to learn discriminative features in aerial images. The framework of D-BiT is illustrated in Fig. 2. We evaluate our model D-BiT extensively on three popular aerial image datasets: Aerial Image Dataset (AID), UC Merced and Event Recognition in Aerial videos (ERA). At test time, we only resize the data and feed it directly into the model. Experimental results show that our D-BiT outperforms SOTA CNN methods with different backbones on these datasets. Notably, D-BiT can provide a significant improvement over the standard ViT model. Further detailed analysis of our method is shown in the ablation studies. The contributions of this work are summarized as follows:

1. We apply ViT to adaptively focus on more discriminative regions in aerial images, called D-BiT, which achieves SOTA performance on several standard benchmarks.
2. To address the fitting dilemma of ViT caused by the lack of trainable aerial images, we design DLSM for the classifier head to enhance ViT to fully learn discriminative features without overfitting the distribution of noisy data, even in the condition of insufficient data volume.
3. Aiming at aerial images with complex backgrounds, we propose a discriminative supervised contrastive loss for a projector head to effectively combine label information and achieve a more compact and reasonable intra-class structure.
4. Extensive experiments show the effectiveness and usefulness of our proposed D-BiT model. Meanwhile, D-BiT can provide a significant improvement over the standard ViT model.

2. Related works

2.1. Aerial image recognition

Aerial image recognition (AIR) is an active research topic in the remote sensing community, which aims to distinguish a predefined semantic category (*e.g.*, “meadow”, “resort”, “railway

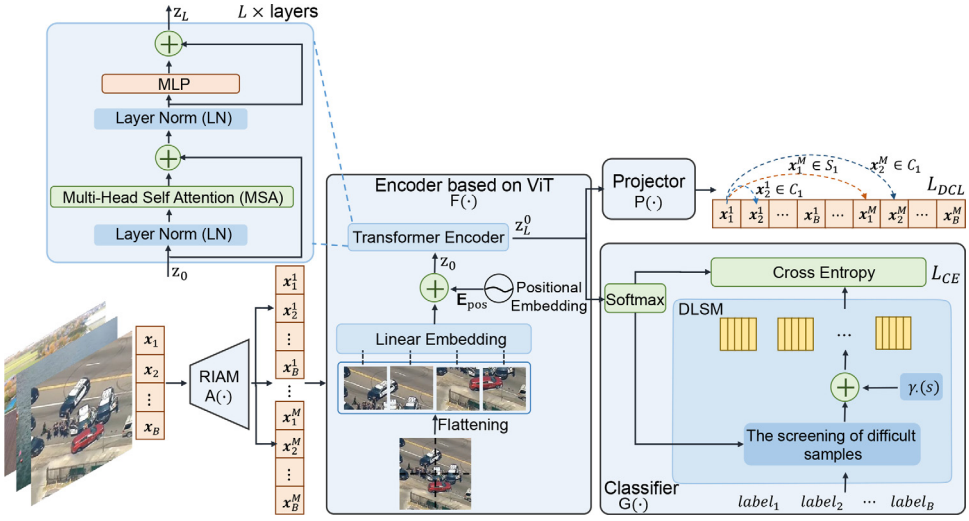


Fig. 2. The architecture of the proposed D-BiT, which includes the Random Image Augmentation Module (RIAM, *e.g.*, A(·)) to increase the diversity of training data, an encoder based on ViT (*e.g.*, F(·)) to extract features, a classifier head (*e.g.*, G(·)) and a projector head (*e.g.*, P(·)). D-BiT enables a dual training mechanism: the discriminative label smoothing module (DLSM) on the classifier head for progressive learning, and the discriminative supervised contrastive loss (DCL) on the projector head to achieve a more compact and reasonable intra-class structure.

station") for each given image according to its visual information [1–3]. Complex backgrounds and various types of objects in aerial images bring huge challenges to recognition. Traditionally, handcrafted features were developed to describe different aerial images, such as SIFT [18], texture information [19] and color [20]. However, these features typically require sufficient expert information to be designed or extracted, and it is difficult for them to depict the high-diversity and non-homogeneous spatial distributions in aerial images.

Recently, deep learning has greatly improved AIR research. Deep learning methods use a multi-stage global feature learning architecture to adaptively learn image features, which can learn more abstract and discriminative semantic features. Among them, some popular convolutional neural network (CNN) architectures have been widely used as the backbone of AIR [21–24], and these CNN-based models [25–29] have also achieved state-of-the-art performance. CNN-based models attempt to explore the global features by gradually expanding the receptive field, while contextual features are ignored. Long-range spatial dependence is a common phenomenon in aerial images, characterized by their complex and compositional structure [1,4,5]. The discriminative features can be distributed in different regions of the aerial image. Therefore, modeling these global relations is crucial for understanding aerial imagery.

2.2. Transformers in computer vision

The vision transformer (ViT) [8] has become increasingly popular for its powerful ability to capture long-range contextual relations, signaling a transition from CNN-based models to ViT-based models. Specifically, ViT is the first attempt to use the pure transformer architecture for image classification. It learns inter-patch representations by dividing and flattening images into a sequence of non-overlapping patches, then using multi-headed self-attention in transformers to learn inter-patch representations. This operation is particularly effective at capturing long-distance dependencies and allows the model to focus on all elements in the input sequence, in sharp contrast to the limited "receptive field" of convolutions. As expected, ViT has achieved competitive performance with CNNs on natural image classification tasks. Built upon the success of ViT, many efforts have been devoted to improving ViT and adapting it to various vision tasks, especially image classification [30–33].

In the remote sensing community, the use of ViT has been limited due to the complexity of large models that requires large-scale datasets for training [8]. For example, in [8], the benefits of ViT only became evident when Google's private 300 million image dataset, JFT-30M, was used for pre-training. The standard solution is to employ transfer learning: Typically, a model is pre-trained on a larger-scale dataset and then fine-tuned for a specific downstream task. In the aerial imaging domain, deep neural network architectures pre-trained on natural images [10] have been reported to generally outperform those trained from scratch [5,11,12], both in terms of final performance and reduced training time. In this work, we extend ViT to recognize aerial images, which can use the self-attention mechanism to capture the global feature information, thereby establishing long-distance dependencies to extract more powerful features.

2.3. Label smoothing

Label smoothing [13] is a regularization technique that has been widely used in many state-of-the-art models, including image classification [34,35], language translation [9], and speech recognition [36]. Label smoothing can effectively restrict the model from overfitting by computing the cross-entropy loss using a weighted mixture of these targets with a uniform noise distribution instead of the "hard" one-hot targets. Notably, the essence of label smoothing at work is that gives true labels a certain degree of fault tolerance. Despite it encouraging the representations of training samples from the same class into tight clusters [14], it also causes the differences between these samples to be lost, which is equivalent to giving up difficult samples to achieve the above aggregation effect. For slightly more complex models, label smoothing easily poses the risk of these model underfitting on small-scale datasets. This dilemma reveals the naive combination of label smoothing and complex models is not suitable for small-scale datasets. The above difficulties are more pronounced on aerial image datasets with noisy and difficult samples.

2.4. Contrastive learning

Contrastive learning has recently become a popular direction in self-supervised learning because of its powerful feature learning ability. The ideas of contrastive learning can go back very far [37,38]. The core idea is to discover some characteristics in the data

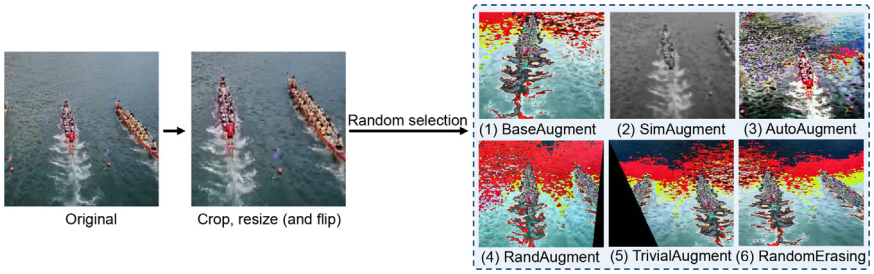


Fig. 3. Pipeline of the Random Image Augmentation Module (RIAM). First, apply a randomly resized crop to the image and randomly flip it horizontally. Then, several augmented image are randomly selected among 6 data augmentation strategies widely used in vision tasks. Each augmentation can transform data stochastically with some internal parameters (*e.g.* rotation degree, noise level).

by learning the difference between the data distribution samples and the noise distribution samples. This approach relies on comparison with noisy data, also known as noise-contrastive estimation (NCE). Later, [39] pointed out the relationship between generative adversarial networks and noise-contrastive estimation. Meanwhile, [16] revealed that contrastive learning is related to mutual information of a pair of positive keys, which is known as InfoNCE. Considering a pair of positive keys $(\tilde{q}, q_0)$ with a large negative keys $\{q_1, \ldots, q_N\}$, where N is the number of negative keys, this kind of non-parametric form van den Oord et al. [16], Wu et al. [40] of contrastive loss can be defined as follows:

$$L_{\text{InfoNCE}} = -\log \frac{\exp(\tilde{q} \cdot q_0/\tau)}{\sum_{i=0}^{N} \exp(\tilde{q} \cdot q_i/\tau)}, \tag{1}$$

where τ is the hyper-parameter for temperature scaling.

However, the standard contrastive loss (*e.g.*, InfoNCE) cannot be used directly for supervised learning because it assumes that each sample in the minibatch belongs to a separate class. To the best of our knowledge, the current state-of-the-art method [17] proposed a supervised contrastive loss to treat all samples with the same label in a minibatch as positive keys, adapting contrastive learning to the fully supervised setting. But aerial images have larger intra-class differences than natural images, and it is inappropriate to regard all samples in the same class as equal positive keys. To this end, we propose a discriminative supervised contrastive loss that not only effectively leverages label information, but also exploits the beneficial properties of contrastive objective functions to discriminately process different augmented images.

3. Proposed method

3.1. Random image augmentation module

ViT, like many deep neural networks, is hard to optimize and can easily overfit if the training data is insufficient. A quick solution to this problem is to apply data augmentation [30] during training. Moreover, the rich data augmentation can also improve the performance of contrastive learning [41]. In this work, based on several image augmentation strategies commonly used in vision tasks, we propose the Random Image Augmentation Module (RIAM), $\mathrm{A}(\cdot)$ to increase the diversity of training data. At the same time, take this to give full play to the ability of contrastive learning.

For an input image, x, $\mathrm{A}(\cdot)$ generates M randomly augmented images $\tilde{x}_j (j \in \{1 \ldots M\})$, each of which represents a different view of the data and thus contains some subset of the information in the original input image x. Denote the training dataset of the aerial images recognition task as $\{(x_i, y_i)\}_{i=1}^{B}$, then it can be extended to a new set $\{(\tilde{x}_i, y_i)\}_{i=1}^{B \times M}$ by $\mathrm{A}(\cdot)$. As shown in Fig. 3, the first stage of augmentation is to apply a randomly resized crop to the image and randomly flip it horizontally. For the second stage, we select 6 ($M \in [0, 6]$) data augmentation strategies widely used in training for current vision tasks: (1) BaseAugment: A lightweight strategy that normalizes only images after the first stage. (2) SimAugment [41]. (3) AutoAugment [42]. (4) RandAugment [43]. (5) TrivialAugment [44]. (6) RandomErasing [45]. The right part of Fig. 3 visualizes the effect of each data augmentation strategy.

3.2. Vision transformer as feature encoder

We denote the vision transformer as the feature encoder, $\mathrm{F}(\cdot)$, which maps an augmented image to a token embedding. In our approach, the M augmented images of each input image are separately input to the same encoder, resulting in M representation vectors. The structure of ViT is mainly composed of three components: a linear layer for patch embedding, a stack of transformer blocks with multi-head self-attention and feed-forward layers for feature encoding, and a linear layer for classification score prediction.

Patch Embedding. Following ViT, we first divide and embed an augmented image $\tilde{\mathbf{x}}_i \in \mathbb{R}^{H \times W \times C}$ in N_P patch tokens $\mathbf{p}_i \in \mathbb{R}^{N_P \times D_E}$, where (H, W) is the resolution of the original image, C is the number of channels, (P, P) is the resolution of each image patch, $N_P = (H \times W)/P^2$ is the resulting number of patches, and D_E is the dimension of embedding. After that, we aggregate the global information by a class token $\mathbf{p}_{\text{cls}} \in \mathbb{R}^{1 \times D_E}$, ViT operates on the patch embedding $\tilde{\mathbf{z}}_0 = [\mathbf{p}_{\text{cls}}, \mathbf{p}_i] \in \mathbb{R}^{(1+N_P) \times D_E}$. A learnable position embedding is added to the patch embeddings to retain positional information as follows:

$$\mathbf{z}_0 = \tilde{\mathbf{z}}_0 + \mathbf{E}_{\text{pos}}, \tag{2}$$

where $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{(1+N_P) \times D_E}$ denotes the position embedding.

The Transformer Encoder. The Transformer encoder contains L layers of multi-head self-attention (MSA) layers and feed-forward multi-layer perceptron (MLP) blocks. Thus the output of the lth layer can be written as follows:

$$\mathbf{z}'_l = \mathrm{MSA}(\mathrm{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1}, \qquad l \in 1, 2, \ldots, L \tag{3}$$

$$\mathbf{z}_l = \mathrm{MLP}(\mathrm{LN}(\mathbf{z}'_l)) + \mathbf{z}'_l, \qquad l \in 1, 2, \ldots, L \tag{4}$$

where $\mathrm{LN}(\cdot)$ denotes the layer normalization operation and $\mathbf{z}_l$ is the encoded image representation. ViT exploits the first token of the last encoder layer $\mathbf{z}_L^0$ as the representation of the global feature and forwards it to a classifier head to obtain the final classification results without considering the potential information stored in the rest tokens.

3.3. Discriminative label smoothing on classifier head

Following ViT, given an augmented image $\tilde{\mathbf{x}}_i$, we still adopt the first token $\mathbf{h}_i = \mathrm{F}(\tilde{\mathbf{x}}_i)$ as the representation of the global feature

and feedforward $\mathbf{h}_i$ into a K-way classifier $\mathrm{G}(\cdot)$, in which K is the number of categories of the aerial image. Denote the parameters of the classifier $\mathrm{G}(\cdot)$ as $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_k, \ldots, \mathbf{w}_K]$, where $\mathbf{w}_k$ corresponds to the parameter for the kth class. $\mathbf{W}$ can be updated by the standard cross-entropy (CE) loss between the output likelihood distribution and one-hot label vector. However, insufficient one-hot representations tend to train models with over-confidence, resulting in poor generalization. At the same time, simply performing label smoothing on small-scale datasets can easily lead to model underfitting to the training data.

In this paper, we propose the Discriminative Label Smoothing Module (DLSM) as an enhancement component of the classifier head to facilitate ViT to learn from simple samples and gradually progress to hard samples and knowledge. With this control, ViT can fully learn discriminative features without overfitting the distribution of noisy data, even in the condition of insufficient data volume. The DLSM mainly consists of a screening module of difficult samples and two smoothing variables, which are detailed as follows.

Firstly, similar to label smoothing, the modified targets can be indicated as:

$$\tilde{y}_k^i = y_k^i(1 - \gamma_\cdot(s)) + \gamma_\cdot(s)/(K-1), \tag{5}$$

where y_k^i is the label indicator of the ith sample, which is "1" for the correct class and "0" for the rest. The parameter $\gamma_\cdot(s)$ is the smoothing variable to control the smoothing weights.

The smoothing variable $\gamma_\cdot(s)$, which is a constant in label smoothing, we replace with two smoothing variables γ_{simple} and γ_{hard} to control the smoothing weights of easy samples and hard samples separately at different training stages:

$$\gamma_{\text{hard}}(s) = \gamma_{\max} - \mathcal{H}_s(\cdot), \tag{6}$$

$$\gamma_{\text{simple}}(s) = (\gamma_{\text{hard}}(s) + \gamma_{\text{bias}}) \times 0.5^{(1+s)}, \tag{7}$$

where $\gamma_{\text{hard}}(s)$ and $\gamma_{\text{simple}}(s)$ are the smoothing variables for hard samples and easy samples, respectively. $s = t/I$, and $t \in \{1 \ldots I\}$ is the number of current training iterations, $\gamma_{\max}$ is the maximum value of smoothing variables for hard samples, and γ_{bias} is the smoothing bias between hard samples and easy samples. $\mathcal{H}_s(\cdot)$ is a smoothing function:

$$\mathcal{H}_s(\tilde{s}) = \tilde{s}^{N+1} \cdot \sum_{n=0}^{N} \mathrm{Comb}_{N+n}^{n} \cdot \mathrm{Comb}_{2\times N+1}^{N-n} \cdot (-\tilde{s})^n, \tag{8}$$

with

$$\tilde{s} = \gamma_{\max} \times s, \tag{9}$$

where Comb$_\cdot^\cdot$ indicates the combinatorial number in mathematics, *i.e.*, Comb_{N+n}^{n} is to calculate the total number of cases where n elements are taken from $N+n$ elements regardless of sorting. N is used to control the rate at which smoothing variables decrease. Its effect under different values is shown in Fig. 4.

Finally, the cross-entropy loss L_{CE} on the classifier head can be formulated as:

$$L_{\text{CE}} = \sum_{i=1}^{B\times M} \sum_{k=1}^{K} -\tilde{y}_k^i \log(p_k^i), \tag{10}$$

where p_k^i is the likelihood the model assigns to the kth class.

$$p_k^i = \exp(\mathbf{h}_i \cdot \mathbf{w}_k) / \sum_{m=1}^{K} \exp(\mathbf{h}_i \cdot \mathbf{w}_m). \tag{11}$$

For the screening of whether the ith sample is difficult, we divide easy and hard samples according to the likelihood the model assigns to the kth class, p_k^i. Intuitively, the likelihood is mostly concentrated in the correct label for simple samples and spread across multiple labels for difficult samples. In addition, two hyperparameters η_1 and η_2 are used to define the screening range of difficult samples. As shown in Fig. 5, consider denoting the maximum and second maximum value of $\{p_k^i\}_{k=1}^K$ as p_m and p_s, respectively. For a sample $(\tilde{x}_i, y_i)$, when p_m is greater than or equal to η_1, and p_s is less than η_2, it is marked as an easy sample, and the rest are divided into hard samples.

3.4. Discriminative supervised contrastive loss on projector head

Previously, we propose the DLSM on the classifier head to facilitate ViT to efficiently learn discriminative features while avoiding training around noisy data even in the condition of insufficient data volume. However, A simple cross-entropy loss is not enough to fully supervise the learning of features since aerial images have greater intra-class variation and inter-class similarity than natural images. And this kind of loss function may still fall short of capturing the intrinsic structure.

Similar to the family of contrastive objective functions, a projector $\mathrm{P}(\cdot)$ [41] is introduced to embed the first token $\mathbf{h}_i$ into a latent metric space as $\mathbf{q}_i$ suitable for computing the contrastive loss. $\mathrm{P}(\cdot)$ is consists of a hidden layer to obtain $\mathbf{q}_i = \mathrm{P}(\mathbf{h}_i) = W^{(2)}\sigma(W^{(1)}\mathbf{h}_i) \in \mathbb{R}^{D_P}$, where σ is a ReLU non-linearity and D_P is the dimension of the vector output by $\mathrm{P}(\cdot)$. Further, $\mathbf{q}_i$ is again normalized to lie on the unit hypersphere, which enables using an inner product to measure distances in the projection space. In particular, the projector is only used for training the supervised contrastive loss. After the training is completed, discard it.

To effectively label information and achieve a more compact and reasonable intra-class structure, we propose the following discriminative supervised contrastive loss (DCL) on the projector head:

$$L_{\text{DCL}} = \frac{-1}{B\times M - 1} \sum_{i=1}^{B\times M} \sum_{j=1}^{B\times M} \mathbb{I}_{i\neq j} \cdot L'_{\text{DCL}}, \tag{12}$$

with

$$L'_{\text{DCL}} = \mathbb{I}\left(\mathbf{q}_j \in C_i\right) \cdot w_i^j + \mathbb{I}\left(\mathbf{q}_j \in S_i\right) \cdot v_i^j, \tag{13}$$

with

$$w_i^j = \log \frac{\exp\left(\mathbf{q}_i \cdot \mathbf{q}_j/\tau\right)}{\sum_m^{B\times M} \mathbb{I}_{m\neq i} \cdot \exp\left(\mathbf{q}_i \cdot \mathbf{q}_m/\tau\right)}, \tag{14}$$

$$v_i^j = \log \frac{\min\left(\exp\left(\mathbf{q}_i \cdot \mathbf{q}_j/\tau\right), \varepsilon\right)}{\sum_m^{B\times M} \mathbb{I}_{m\neq i} \cdot \exp\left(\mathbf{q}_i \cdot \mathbf{q}_m/\tau\right)}, \tag{15}$$

where $B \times M$ is the size of candidate keys, $\mathbb{I}_\cdot \in \{0, 1\}$ is an indicator function that values "1" if only if the input condition holds. Among candidate keys of the same class as the ith sample, S_i indicates the set of samples augmented by the same image, and the rest are aggregated into C_i. $\varepsilon (1 \geq \varepsilon > 0)$ is the similarity threshold and $\tau > 0$ is a scalar temperature parameter.

By encouraging samples in the training dataset to be close to their corresponding class prototypes and further mining the internal information of the data, L_{DCL} tends to achieve a more compact and reasonable intra-class structure. As a result, clusters of points belonging to the same class are pulled together in the embedding space, and points augmented from the same image are more concentrated than other points of the same class, while clusters of samples from different classes are pushed apart.

3.5. Total loss function

Finally, the D-BiT framework, a novel approach, is proposed to facilitate ViT to better extract discriminative semantic features from aerial images. As illustrated in Fig. 2, D-BiT includes the

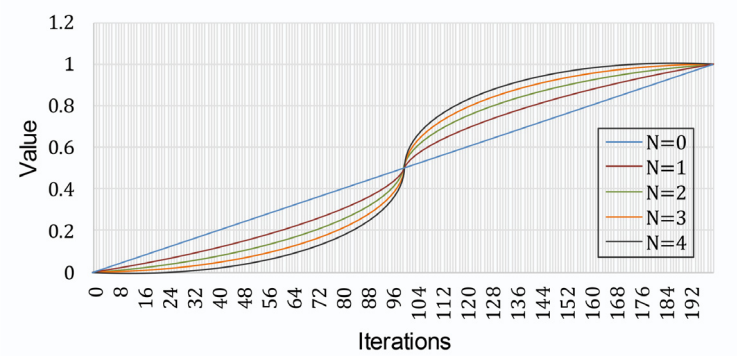


Fig. 4. Illustration of the effect of different values of N on the changing trend.

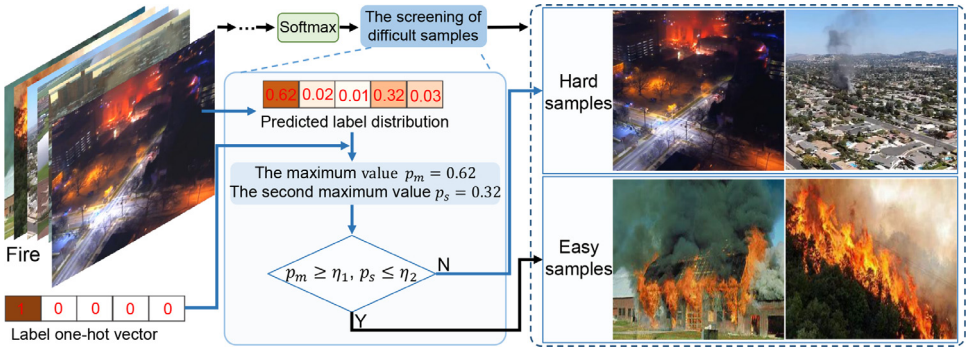


Fig. 5. The screening of difficult samples. Divide easy and hard samples based on the likelihood the model assigns to the sample. Specifically, two hyperparameters η_1 and η_2 are used to define the screening range of difficult samples.

DLSM for the classifier head and a discriminative supervised contrastive loss for the projector head. The overall loss function of D-BiT can be formulated as follows:

$$L = L_{\mathrm{CE}} + \beta \times L_{\mathrm{DCL}}, \tag{16}$$

where β is a coefficient to trade-off L_{CE} and L_{DCL}.

4. Experiments

4.1. Datasets and evaluation metrics

Datasets. To assess the effectiveness of our proposed D-BiT, we choose 3 benchmark datasets: The ERA dataset [3] is an aerial image dataset for event recognition that contains 2864 samples with 25 classes. To our best knowledge, ERA is by far the largest dataset for event recognition in aerial images. Different from remote sensing scene datasets, ERA has the challenges of greater intra-class variations and imbalanced distribution across different classes. Its main pixels belong to the background, so it is difficult to find discriminative semantic features to distinguish events in similar backgrounds. The AID dataset [1] contains 30 categories, each with 200 to 400 images. The UC Merced dataset [2] contains 21 categories, and each category contains 100 images, all of which are extracted from the U.S Geological Survey (USGS) National Map Urban Aera Imagery. In addition, following previous works [1,2,11,46–49], we adopt two different settings for UC Merced and AID datasets for fair comparison. For the UC Merced dataset, the training rate is set to 50% and 80%. For AID dataset, the training rate is set to 20% and 50%.

Evaluation Metrics. Following the previous works, we use the overall accuracy (OA) to evaluate the model performance. In addition, we further present the average accuracy (AA) on the ERA dataset.

4.2. Implementation details

The proposed D-BiT is implemented on the MindSpore and Pytorch platforms and trained with four Nvidia TU102 GPUs. All input images are resized to 224×224 (random cropping for training and testing). We load intermediate weights from the official ViT-B/16 model pre-trained on ImageNet21k. The mini-batch size is set to 10, and the SGD optimizer is employed with a momentum of 0.9. The learning rate is initialized as 0.001. We adopt cosine annealing as the scheduler of an optimizer.

The size M of randomly augmented images in the RIAM is set to 4. In the screening of hard samples, two parameters η_1 and η_2 are used to define the scope of screening difficult samples. In order to fit the conventional situation [15], we uniformly set η_1 and η_2 to 0.8 and 0.2, respectively. $\gamma_{\max}$ in Eq. (6) is set as 0.8 except 0.5 for the UC Merced dataset. γ_{bias} in Eq. (7) is set as 0.2. The parameter N in Eq. (8) is used to control the rate at which smoothing variables decrease. It can be shown from Fig. 10 that the proposed method can obtain the best OA when N is set as 1. Hence, the parameter N is set to 0 except 1 for the ERA dataset. Following [41], the dimension D_P of the vector output by $\mathrm{P}(\cdot)$ is set to 128. OA varies with different values of the parameters τ and ε in Eq. (15) on ERA and UCM datasets, shown in Table 6 and Fig. 7, respectively. It can be seen from Table 6 and Fig. 7 that the proposed D-BiT approach can obtain the best OA when τ and ε are set to 0.3 and 0.8, respectively. Hence, the parameters τ and ε in the proposed D-BiT approach are set to 0.3 and 0.8, respectively. The parameter β in Eq. (16) is adopted to control the weight of DCL. It requires optimal selection for different methods. Empirically, we find 0.5 to be the best value of β in our experiments. Hence, β is set to 0.5. We repeat the experiments three times, and the average results and their standard deviations are reported. In addition, in the various ablation experiments in Section 4.4, we do not nor-

Table 1
Performance comparison (%) with baseline models on ERA dataset.

Model	OA	AA
Swim-T [31]	64.2	65.3
NASNet-L [35]	60.2	63.2
DenseNet-121 [24]	61.7	63.5
DenseNet-169 [24]	60.6	64.4
DenseNet-201 [24]	62.3	61.7
MobileNet [50]	61.3	61.8
Inception-v3 [34]	62.0	64.6
ResNet-50 [23]	57.3	59.0
ResNet-101 [23]	55.3	59.2
ResNet-152 [23]	56.1	59.0
VGG-16 [22]	51.9	55.5
VGG-19 [22]	49.7	51.3
ViT-B/16 [8]	63.6	64.2
D-BiT (ours)	68.6	69.5

Table 2
Comparisons of OA (%) with state-of-the-art methods on AID dataset.

Method	Training ratios	
	20%	50%
PiT-S [32]	90.51±0.57	94.17±0.36
PVT-Medium [33]	92.13±0.45	95.28±0.23
T2T-ViT-12 [54]	90.09±0.08	93.82±0.55
MG-CAP [47]	93.34±0.18	96.12±0.12
MSCP [55]	91.52±0.21	94.42±0.17
GBNet [56]	92.20±0.23	95.48±0.12
MS2AP [48]	92.19±0.22	94.82±0.20
MobileNet [57]	88.53±0.17	90.91±0.18
EfficientNet [58]	86.56±0.17	88.35±0.16
Two-Stream Deep Fusion Framework [59]	92.32±0.41	94.58±0.25
DenseNet121 [52]	93.76±0.23	94.73±0.26
DenseNet169 [52]	92.43±0.36	94.17±0.33
TEX-Net-LF [51]	93.81±0.12	95.73±0.16
Two-Stage Deep Feature Fusion	-	94.65±0.33
GoogLeNet [1]	83.44±0.40	86.39±0.55
ResNet50 [53]	92.39±0.15	94.69±0.19
InceptionV3 [53]	93.27±0.17	95.07±0.22
CaffeNet [1]	86.86±0.47	89.53±0.31
VGG-VD-16 [1]	86.59±0.29	89.64±0.36
VGG19 [53]	87.73±0.25	91.71±0.42
Fusion by addition [60]	-	91.87±0.36
ViT-B/16 [8]	92.12±0.21	94.62±0.30
D-BiT (ours)	94.21±0.20	96.59±0.24

malize the test data and directly feed the resized images into the trained model for analysis.

4.3. Compared methods

Results on ERA. To our best knowledge, ERA [3] is by far the largest dataset for event recognition in aerial images. Different from remote sensing scene datasets, ERA has the challenges of greater intra-class variations and imbalanced distribution across different classes. Its main pixels belong to the background, so it is difficult to find discriminative semantic features to distinguish events in similar backgrounds. As shown in Table 1, our method outperforms all previous methods on ERA. Compared to the best result up to now, our D-BiT improves OA and AA by 4.4% and 4.2%, respectively. Compared to our base framework ViT, OA and AA are improved from 63.6% to 68.6% and form 64.2% to 69.5%, respectively. This strongly demonstrates that our method can significantly improve the accuracy of ViT for aerial image recognition.

Results on AID. The performance comparison on AID [1] is summarized in Table 2. The "-" in Table 2 means that the model did not complete the experiment under 20% for training. The images in AID are multisource images, which is more challenging than the recognition of single-source images. Compared with the UC Merced dataset [2], the AID dataset has greater inter-class similarities and intra-class variations, and a larger scale. When the training ratio is 20%, D-BiT achieves 94.21%, 0.87 higher than MG-CAP [47], 0.4% higher than TEX-Net-LF [51], 0.45% higher than DenseNet121 [52], 1.82% higher than ResNet50 [53] and 2.08% higher than PVT-Medium [33]. When the training ratio is 50%, our method achieves 96.59%, outperforming most existing methods. The experimental results prove that D-BiT also has an outstanding ability to understand aerial images. Bi et al. [48] proposed a multi-scale staking attention pooling (MS2AP) to strengthen the local semantic representation and multi-scale feature representation. D-BiT is 1.77% higher than MS2AP, which not only demonstrates the effectiveness of our method but also reveals the superiority of the self-attention mechanism. Although transformer-based methods are general-purposed, their behavior is still assertive due to the powerful feature learning ability of the transformer model. Meanwhile, D-BiT consistently gets 2.09% and 1.97% improvement compared to the standard ViT model. These encouraging results confirm that our method is helpful to the AID dataset.

Table 3
Comparisons of OA (%) with state-of-the-art methods on UC Merced dataset.

Method	Training ratios	
	50%	80%
PiT-S [32]	95.83±0.39	98.33±0.50
PVT-Medium [33]	96.27±0.42	98.48±0.49
T2T-ViT-12 [54]	95.68±0.61	97.81±0.49
Swim-T [31]	92.24±0.23	93.38±0.13
DSENet [11]	96.19±0.13	99.14±0.22
MIDCNet [46]	94.93±0.51	97.00±0.49
MSCP [55]	-	98.36±0.58
MobileNet [57]	92.17±0.13	96.33±0.15
EfficientNet [58]	90.91±0.16	94.37±0.14
Fine-tune MobileNet V2 [63]	97.88±0.31	98.13±0.33
SRSCNN [64]	-	95.57
Two-Stream Deep Fusion Framework [59]	96.97±0.75	98.02±1.03
TEX-Net-LF [64]	96.91±0.36	97.72±0.54
D-CNN with AlexNet [65]	-	96.67±0.10
D-CNN with GoogleNet [65]	-	97.07±0.12
D-CNN with VGGNet-16 [65]	-	98.93±0.10
Fusion by addition [60]	-	97.42±1.79
Pre-trained-AlexNet-SPP-SS [62]	-	96.67±0.94
CaffeNet [1]	93.98±0.67	95.02±0.81
VGG-VD-16 [1]	94.14±0.69	95.21±1.20
GoogLeNet [1]	92.70±0.60	94.31±0.89
Integrating multilayer features of CNN [61]	-	98.81±0.38
ViT-B/16 [8]	95.80±0.45	98.10±0.33
D-BiT (ours)	97.71±0.31	99.05±0.21

Results on UC Merced. The results of experiments on UC Merced dataset [2] are shown in Table 3. The "-" in Table 3 means that the model did not complete the experiment under 50% for training. Clearly, the most recent methods achieve quite similar results on this dataset. When the training rate is 80%, our method can achieve 99.05%, surpassing most existing methods and comparable to CNN-based methods dedicated to remote sensing scene classification. This indicates that higher accuracy can also be achieved by ViT-based methods. When the training ratio is 50%, the overall accuracy of our method achieves 97.71%. We argue that the amount of training data for UC Merced may be too small, and the ViT-based methods are very sensitive to the amount of data. Li et al. [61] presents a fusion strategy for integrating multilayer features of a pre-trained CNN model for scene classification, but it is still 0.95% lower than our method. Han et al. [62] utilizes Pyramid pooling modules to exploit global information, but our method outperforms it by 2.38%, indicating that the self-attention mechanism is more effective than pyramid pooling modules. Notably, our D-BiT consistently gets 1.91% and 0.95% improvement compared to the standard ViT model.

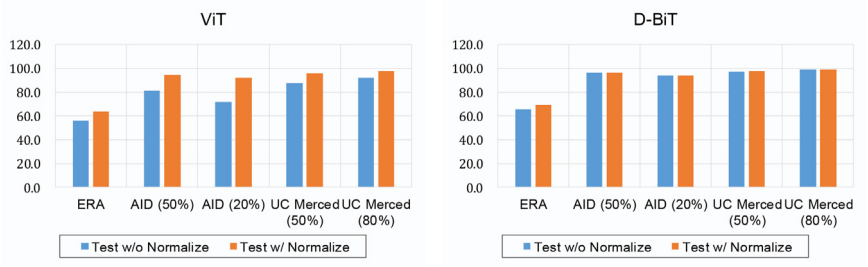


Fig. 6. Evaluation of the impact (%) of whether the test set is normalized or not on the stability of the model performance.

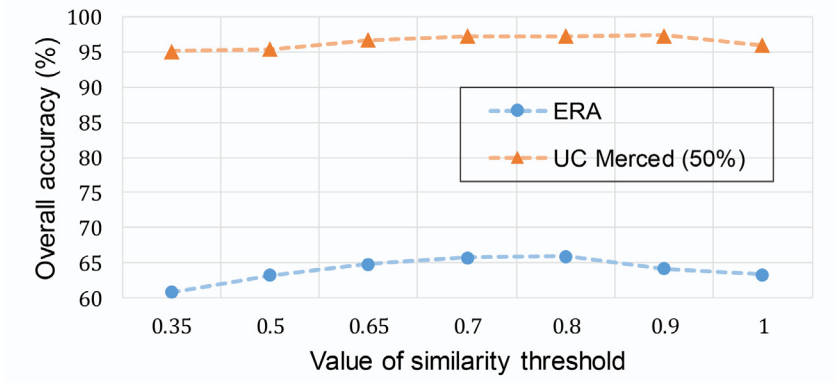


Fig. 7. Sensitivity analysis (%) of similarity threshold ε on ERA and UC Merced (50%).

Table 4
Evaluation (%) of RIAM, DLSM and DCL on ERA and UC Merced (50%).

RIAM	DLSM	DCL	ERA	UC Merced (50%)
			56.00	88.00
	√		59.20	90.32
√			62.44	94.85
√		√	64.52	95.48
√	√	√	65.90	97.43

Table 5
Comparison between our DCL and SCL on ERA and UC Merced (50%).

Loss	RIAM	DLSM	ERA	UC Merced (50%)
SCL	√	√	63.34	96.12
	√		62.27	94.25
DCL	√	√	65.90	97.43
	√		64.52	95.48

4.4. Ablation studies and further discussion

Effect of test set normalization. During data preprocessing, normalization can speed up the convergence of the model to normalize the dataset. Forcing the distribution of the training set to the test set will lead to data leakage, so the training and test sets need to be normalized separately. The performance of the model is reflected in its performance on unknown new data, and the information of the test set must not be introduced when training the model. We find that normalization on the test set has a huge impact on the performance of ViT while having a small impact on the performance of our model. As shown in the Fig. 6, when the test set is not normalized, the performance of ViT decreases sharply, while our model is stable. This reflects that our model is more generalizable.

The effectiveness of components. To validate the proposed modules in our D-BiT, an ablation study is designed to investigate the effects of RIAM, DLSM, and DCL, as shown in Table 4. To efficiently show results, a tuple (a,b) is used where a and b denote the performance on ERA and UC Merced datasets, respectively. The baseline strategy (the first row) means there is no RIAM, no DLSM, and no DCL, *i.e.*, the input images are directly fed into the standard ViT encoder. By combining the DLSM, the overall accuracy is improved form to (56.0%, 88.0%) to (59.20%, 90.32%). This indicates the effectiveness of discriminative label smoothing. Compared with the baseline, RIAM improves performance by (6.44%, 6.85%) and further gains (2.08%, 0.63%) improvement by adding DCL. We argue that this is because DCL can help the model to learn the intrinsic structure of training data, effectively enlarging the representation distance between similar classes. In particular, DLSM can further improve the performance, which achieves the higher performance of (65.90%, 97.43%), improved by (1.38%, 1.95%). We believe that this is because DLSM makes the model learn gradually from easy to difficult, which better promotes the stability of model training.

Comparison between discriminative supervised contrastive loss (DCL) and supervised contrastive loss (SCL). DCL effectively combines label information, and at the same time, it discriminatively processes samples augmented by different images in the same class. This is why we call it a discriminative supervised contrastive loss. In contrast, supervised contrastive loss (SCL) [17] directly treats all samples with the same label in a minibatch as positive keys, which is not suitable for aerial images with large interclass differences. To verify our hypothesis, we perform experiments with SCL and our proposed DCL on ERA and UC Merced dataset. We only replace DCL with SCL and experiment separately with or without DLSM. Other settings are the default as described in 4.2. The experimental results are shown in Table 5. We can observe that our DCL is higher than SCL with or without DLSM. This indicates that our hypothesis is valid, that for aerial images, training data augmented by different images in the same class should be treated further discriminatively.

Effect of the temperature parameter τ. The result of different setting of the scalar temperature τ in Eq. (15) is shown in Table 6.

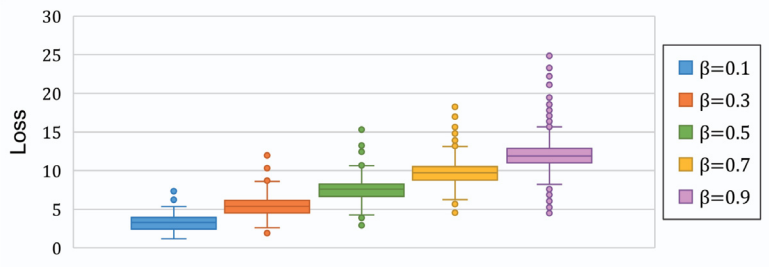


Fig. 8. Comparison of the variability of overall loss to changes in trade-off coefficient β on ERA.

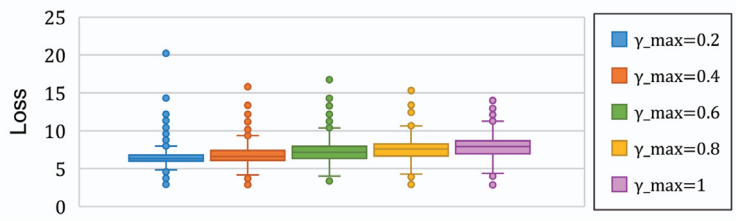


Fig. 9. Comparison of the variability of overall loss to changes in the maximum value $\gamma_{\max}$ of smoothing variables for hard samples on UC Merced (50%) dataset.

Table 6
Ablation study on the value of temperature parameter τ on ERA and UC Merced (50%).

Method	Value of τ	ERA	UC Merced (50%)
D-BiT	0.1	55.37	88.04
	0.2	63.44	94.85
	0.3	65.90	97.43
	0.4	61.34	94.95

We find that a smaller value of τ leads to training signals dominated by difficult negative samples while overemphasizing separation from difficult negative samples will destroy the learned latent semantics. However, a higher value of τ prevents the model from learning enough information to enlarge the distance of difficult negative samples. Empirically, we find 0.3 to be the best value of τ in our experiments.

Effect of similarity threshold ε. In Fig. 7, we report the sensitivity of the similarity threshold ε in Eq. (15). The value of ε reflects the tolerance for similarity in the representation of samples augmented by different images in the same class. A larger value of ε will lead the model to ignore differences between different augmented samples in the same class. And a smaller value of ε will cause the model to learn the wrong label information. Empirically, we find 0.8 to be the best value of ε in our experiments.

Effect of the trade-off coefficient $\beta$for loss functions. In Fig. 8, we compare the stability of the loss drop during model training for different values of β in Eq. (16). We see significantly more outliers in the output of the loss when increasing the value of β. While we do not have a theoretical explanation of this behavior, we conjecture that this is due to the inherent training instability of transformers in the self-supervised learning framework. As the proportion of contrastive loss increases, D-BiT tends to self-supervised learning mode, leading to an increase in the instability of training. Instability is the most important problem that leads to loss of accuracy. This result suggests that there is a trade-off between stability and the beneficial properties of contrastive loss. Empirically, we find 0.5 to be the best value of β in our experiments.

Effect of the maximum value $\gamma_{\max}$ of smoothing variables for hard samples. Fig. 9 shows the effect of different values of $\gamma_{\max}$ in Eq. (6) on the stability of the loss value drop during model training. We find that both too large and too small values of $\gamma_{\max}$ can affect the stability of model training. As the value of $\gamma_{\max}$ increases, the stability of model training improves. However, if the value of $\gamma_{\max}$ is too large, the model converges too slowly and requires a longer total number of iterations, so it should be appropriate.

Effect of smoothing rate N. The comparison result of different settings of smoothing rate N in Eq. (8) is shown in Fig. 10. N is used to control the rate at which smoothing variables decrease. The larger the value of N, the faster the smoothing will be in the middle stage and the slower the smoothing will be in the early and late stages.

Algorithm 1 D-BiT algorithm.

Input: ~~
Training dataset $\{(x_i, y_i)\}_{i=1}^{B}$.
Output: ~~
All weight parameters **W** of D-BiT.
All bias parameters **V** of D-BiT.
Initialization: ~~
The initial weights of the encoder adopt the weights pretrained on ImageNet21k, and the projetor and classifier head are randomly sampled by truncated distribution.
Repeat: ~~
1: For an input image x, A$(\cdot)$ generates M randomly augmented images, and training dataset can be extended to a new set $\{(\tilde{x}_i, y_i)\}_{i=1}^{B\times M}$ by A$(\cdot)$;
2: Use vision transformer to encode images x_i into representation vectors $\mathbf{h}_i$,and feedforward $\mathbf{h}_i$ into the classifier G$(\cdot)$ and projector P$(\cdot)$;
3: Divide easy and hard samples according to the likelihood the model assigns to the kth class;
4: Calculate the modified targets $\tilde{y}^i$ according to (5), (6), (7) and (8);
5: Calculate the cross-entropy loss L_{CE} and discriminative supervised contrastive loss L_{DCL} according to (10) and (12);
6: Leverage L_{CE} and L_{DCL} to calculate L according to (16);
7: Update the parameters **W** and **V** by utilizing SGD.
Until: a fixed number of iterations ~~
Return: **W**, **V** ~~

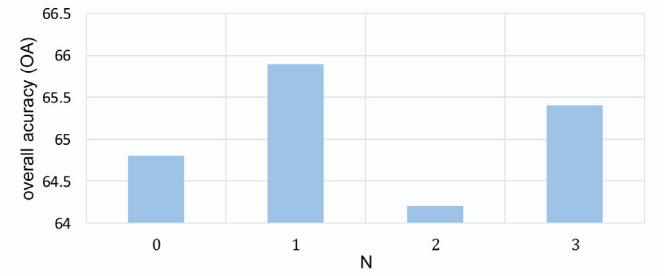


Fig. 10. Evaluation of the impact (%) of different settings of smoothing rate N on ERA dataset.

5. Conclusion and future work

In this paper, we proposed a new framework based on ViT for the aerial image recognition task, called D-BiT. D-BiT aims to recognize more diverse aerial images, rather than being limited to a specific task such as remote sensing scene classification. First, Discriminative Label Smoothing Module (DLSM) is designed to enhance ViT to fully learn discriminative features without overfitting the distribution of noisy data, even in the condition of insufficient data volume. Then, a novel contrastive loss is designed to effectively combine label information and achieve a more compact and reasonable intra-class structure. We conducted extensive experiments to evaluate D-BiT on three popular aerial image datasets. Experimental results demonstrate that our approach outperforms most existing CNN methods with different backbones, and verify that D-BiT can significantly improve the accuracy of ViT in recognizing aerial images. Our method does not reduce the number of parameters in ViT, and the high complexity in time and space limits the application potential of ViT-based models. Therefore, we will further design a more lightweight ViT-based model or a CNN-ViT hybrid model for AIR in future work. In addition, a reasonable label distribution should more truly reflect the correlation among categories, and the self-attention mechanism of ViT can capture rich salient features, which may be a direction worth considering.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Shengwu Xiong reports was provided by Wuhan University of Technology. Shengwu Xiong reports a relationship with Wuhan University of Technology that includes: funding grants.

CRediT authorship contribution statement

Yichen Zhao: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Writing – review & editing. **Yaxiong Chen:** Methodology, Validation, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Xiongbo Lu:** Data curation, Writing – review & editing. **Lei Zhou:** Resources, Visualization. **Shengwu Xiong:** Supervision, Funding acquisition.

Data availability

The authors do not have permission to share data.

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